Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - II

Course Code: BCA104-2 **Max Marks: 100**

Course Name: DISCRETE MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Demonstrate a working knowledge of set notation and elementary set theory and recognize the connection between set operations and logic.

CO2: Apply the different properties of injections, surjections, bijections, compositions, and inverse functions.

CO3: Understand different types of matrices and operations among matrices.

CO4: Determine when a relation is reflexive, symmetric, antisymmetric, transitive, apply the properties of equivalence relations and partial orderings, and explain the connection between equivalence relations.

CO5: Effective express the elementary concepts and results of Number theory.

Section A **3*8=24**

1)	Discuss how matrix representation can be used to determine if a given relation on a set is symmetric. Provide an example using a 3x3 relation matrix.	**3**	(CO3, L2)
2) a)	Examine and prove whether the Cartesian product of two countable sets A and B is also countable.	**3**	(CO2, L4)
	OR		
b)	Examine and differentiate between injective, surjective, and bijective functions with examples from the domain {1, 2, 3} to codomain {a, b, c}. Also, determine if a given function is a bijection by analyzing its matrix representation.	**3**	(CO2, L4)
3)	Determine whether the function : Z → Z defined by () = 3 + 2 is injective and surjective. Justify your answer.	**3**	(CO5, L3)
4)	Examine and differentiate between injective, surjective, and bijective functions using specific examples with domain and codomain as sets of integers.	**3**	(CO2, L4)
5) a)	Discuss how you would use the concept of a function to determine if a given relation on a set A is injective and provide an example using specific elements from A.	**3**	(CO2, L2)
	OR		
b)	Discuss how matrix representation can be used to determine if two sets have the same cardinality.	**3**	(CO2, L2)
6)	Determine the smallest positive integer that is congruent to 5 modulo 7 and also divisible by 4.	**3**	(CO1, L3)

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7) a)	Analyze how modular arithmetic can be applied to determine if two given integers are coprime without directly computing their greatest common divisor (GCD). Provide an example using your analysis.	**3**	(CO1, L4)
	OR		
b)	Examine and prove whether the set of all finite subsets of natural numbers is countable.	**3**	(CO1, L4)
8)	Examine and differentiate between a symmetric matrix and a skew-symmetric matrix by providing an example of each. Also, analyze how these matrices can represent relations in discrete mathematics.	**3**	(CO3, L4)
	Section B		**8*7=56**
9)	Discuss how to prove that a given set is countable by establishing a one-to-one correspondence with the set of natural numbers. Then, apply this method to show that the set of all rational numbers (fractions) between 0 and 1 is countable.	**8**	(CO1, L2)
10) a)	Consider the functions f: A → B and g: B → C defined by f(x) = 2x + 1 and g(y) = y^2 - 4. Determine if these functions are injective, surjective, or bijective when their domains are the set of all integers Z. Discuss your findings using examples to support your arguments. Then, calculate the composition g f: A → C and determine its properties based on the results from part (i).	**8**	(CO2, L2)
	OR		
b)	Discuss the properties of injections, surjections, and bijections with respect to Cartesian products. Given two non-empty sets A and B, where |A| = 3 and |B| = 4, illustrate these concepts by defining functions f: A × B → B and g: B × A → A, and determine if they are injections, surjections, or bijections.	**8**	(CO2, L2)
11)	Construct two different 4 × 4 matrices A and B that represent symmetric relations on a set S = {1, 2, 3, 4}. Then, calculate the matrix C representing the union of these relations (i.e., =). Demonstrate whether the resulting relation represented by C is symmetric. Finally, determine if the composite relation D = A • B contains any ordered pairs (i, j) where i < j.	**8**	(CO3, L3)
12) a)	Explain how to use modular arithmetic to determine whether a given integer is congruent to another modulo n. Discuss with specific examples involving addition, multiplication, and the application of the properties of congruence.	**8**	(CO1, L2)
	OR		
b)	Discuss and illustrate with examples how to prove whether the union of two given sets is countable. Specifically, consider the following cases and justify your answers: (i) When both sets are finite; (ii) When one set is finite and the other is infinite but countable; (iii) When both sets are infinite and countable.	**8**	(CO1, L2)
13)	Analyze how to determine whether a given relation is reflexive, symmetric, antisymmetric, or transitive on the set of integers modulo 7 using an example relation R defined as R = {(x, y) | x ≡ y (mod 7)}. Then, differentiate between this relation and another relation S defined as S = {(x, y) | 3x ≡ 4y (mod 7)} in terms of the properties mentioned. Finally, prove whether either relation is an equivalence relation or a partial ordering on the set of integers modulo 7.	**8**	(CO4, L4)

14) a)	Consider two matrices A and B defined as follows: A = [1 0 1; 0 1 0; 1 0 1] B = [1 1 0; 1 0 1; 0 1 1] Determine the following operations and justify your answers with appropriate reasoning: (i) Compute A + B (element-wise addition). Is the resulting matrix symmetric? Justify. (ii) Determine AB, the product of matrices A and B. Describe the process used to obtain this result. (iii) Find the transpose of matrix B. Explain whether the transpose operation affects the symmetry property of a matrix.	**8**	(CO3, L3)
	OR		
b)	Construct two different 4 × 4 matrices A and B such that the product AB results in a matrix with only even entries, but BA does not have this property. Demonstrate your solution by explicitly calculating both products. (i) Write down the matrices A and B. (ii) Calculate the product AB and show that it has only even entries. (iii) Calculate the product BA and show an example where one of its entries is odd.	**8**	(CO3, L3)
15)	Consider two functions f: A → B and g: C → D where A = {1, 2, 3}, B = {4, 5, 6}, C = {7, 8, 9}, and D = {10, 11, 12}. Determine whether the following statements are true or false. If false, provide a counterexample to disprove each statement. (i) The composition g f: A → D is an injection. (ii) The function g: C → D is a surjection. (iii) There exists a bijection between set A and set C.	**8**	(CO2, L3)
	Section C		**10*2=20**
16)	Determine whether the relation R defined on the set A = {1, 2, 3, 4, 5} by xRy if and only if x ≡ y (mod 2) is reflexive, symmetric, antisymmetric, or transitive. If it is an equivalence relation, find all the equivalence classes. If not, specify which property fails and provide a counterexample. Additionally, show that R can be represented as a partition of A.	**10**	(CO4, L3)
17) a)	Examine and differentiate between a symmetric relation and an antisymmetric relation using examples from a set of integers under the relation 'divides'. Additionally, analyze how these properties connect to equivalence relations and partial orderings in discrete mathematics. Provide specific counter-examples where necessary.	**10**	(CO4, L4)
	OR		
b)	Analyze and differentiate between a symmetric relation and an antisymmetric relation by proving or disproving given relations on specific sets. Consider two non-empty finite sets A = {1, 2, 3} and B = {a, b, c}. Define the following relations R and S on set A: R = {(1, 1), (2, 2), (3, 3)} S = {(1, 2), (2, 1), (2, 3), (3, 2)}. (i) Prove whether relation R is symmetric or not. Provide a counterexample if it is not symmetric. (ii) Determine if relation S is antisymmetric. Justify your answer with necessary examples and counterexamples. (iii) Explain the differences between symmetric and antisymmetric relations based on your findings.	**10**	(CO4, L4)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create